Instruction:

1. **Copy all the output images and paste into the ONE docx file, then save it as pdf.**
2. **Submit the pdf file together with all the m-files.**
3. **Name your M-files as: StudentID_Quiz_1_Q#.m, where # is the question number. For example, MAT1234567_Quiz_1_Q1.m. Take note that no spaces are allowed in the file name.**

Question 1 (25 marks)

Given a piecewise function $f(x)$ as follows:

$$f(x)=\begin{cases} 2, & -\pi \le x \le \dfrac{\pi}{2} \\ \dfrac{1}{\sin(x)}, & \dfrac{\pi}{2} < x < 2\pi \end{cases}$$

Plot the graph $y = f(x)$. Title the graph as "StudentID: Graph of a piecewise defined function" in Times New Roman font.

Remark:

a) Label x-axis in unit of π.
b) Plot dashed-line as asymptote line (if any).

Question 2 (25 marks)

Produce a graph as shown in Figure 1. Title the image as "StudentID: Area bounded by three curves" in Times New Roman font.

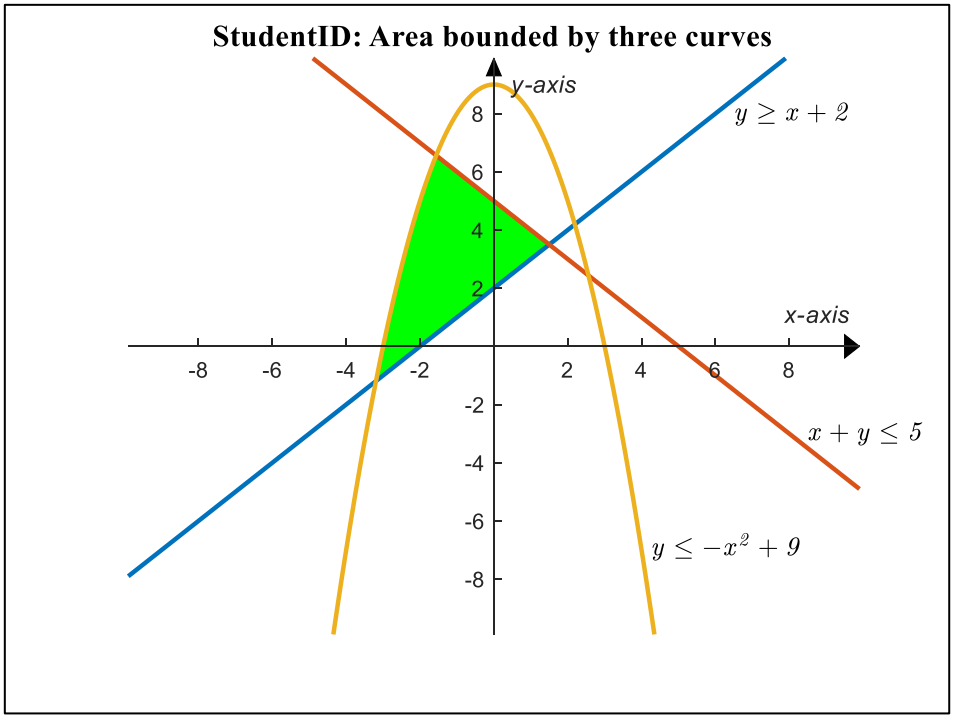


Figure 1